brca12-wgs: a reproducible Nextflow pipeline for *BRCA1/BRCA2* germline analysis from whole-genome sequencing on GRCh38

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Software: https://github.com/pjlabsdev/brca12-wgs · DOI: 10.5281/zenodo.24173638 (version 1.0.1) · Licence: MIT

Summary. Germline pathogenic variants in *BRCA1* and *BRCA2* cause hereditary breast and ovarian cancer syndrome. The relevant spectrum spans single-nucleotide variants, small insertions and deletions, exon-level copy-number changes and larger rearrangements, yet many open BRCA workflows target amplicon panels on GRCh37 and report small variants only. *brca12-wgs* is a Nextflow pipeline that takes paired-end FASTQ files or a BAM/CRAM from whole-genome sequencing and produces a structured BRCA1/BRCA2 report on GRCh38. It combines BWA-MEM, GATK HaplotypeCaller or DeepVariant, Manta, CNVkit with per-exon depth ratios, Ensembl VEP with MANE Select transcripts, ClinVar and mosdepth, and lists automated hints on single ACMG/AMP criteria following the ClinGen ENIGMA specification without assigning a classification. We measured performance on four kinds of data: Genome in a Bottle truth sets (364 of 364 SNVs and 67 of 69 indels), ten 1000 Genomes carriers of pathogenic variants (10 of 10 detected), 185 small and 12 large variants spiked into real reads, and real Alu insertions and deletions in 1000 Genomes samples. Real Alu insertions were called in only 4 of 11 carriers, but an insertion signature from clipped read ends, added for this reason, was present in 11 of 11. Manta reported all 12 spiked large events, whereas single-sample CNVkit segmentation reported 3 of 12 after a correction made during this work (0 before). The software is for research use; results require confirmation by a clinically validated method.

Availability. Source code, test data, validation scripts and results are available under the MIT licence at https://github.com/pjlabsdev/brca12-wgs.

Highlights

- One command from WGS reads to a BRCA1/BRCA2 report with small variants, structural variants, copy number and exon coverage on GRCh38.
- 10 of 10 real carriers of pathogenic variants from the 1000 Genomes Project were detected with the correct genotype.
- 185 of 185 spiked SNVs and indels up to 40 bp and 12 of 12 spiked deletions, duplications and an Alu insertion were reported.
- Real Alu insertions were called in 4 of 11 carriers; a read-end signature added for this reason was present in 11 of 11 and in 0 of 47 non-carrier evaluations.
- The validation exposed and fixed three defects, including missed copy-number segments.
- Research use only. No classification is assigned.

Research use only. brca12-wgs is not a medical device and has not been clinically validated. Its output must not be used for diagnosis, risk assessment or treatment decisions without confirmation in an accredited laboratory (for example Sanger sequencing for small variants and MLPA for copy-number changes). ACMG/AMP entries are automated hints on individual criteria, not variant classifications. A result without findings does not exclude a hereditary BRCA1/BRCA2 predisposition.

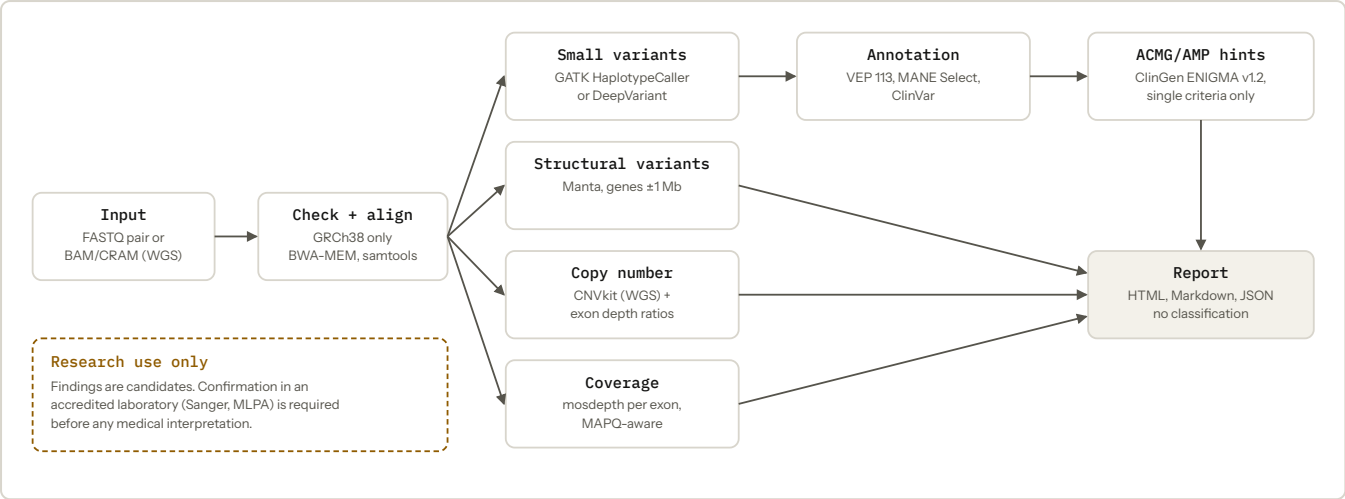


Figure 1. Workflow. Reads are checked against GRCh38 and aligned. Four evidence layers are computed independently and kept separate in the report. Hints on single ACMG/AMP criteria are attached to annotated small variants; no combination rule is applied.

1 Introduction

Pathogenic germline variants in the tumour suppressor genes *BRCA1* (chr17, minus strand) and *BRCA2* (chr13, plus strand) substantially increase lifetime risks of breast, ovarian, prostate and pancreatic cancer [1]. Most pathogenic alleles are truncating SNVs and small indels, but large genomic rearrangements such as single- or multi-exon deletions and duplications account for a relevant fraction of findings, particularly in *BRCA1*, which is rich in Alu repeats and has a partially duplicated pseudogene (*BRCA1P1*) near its 5′ end [2]. Insertions of mobile elements are rare but include a founder variant, the Alu insertion c.156_157insAlu in *BRCA2* exon 3 [3, 4].

Whole-genome sequencing (WGS) captures all of these variant classes in a single assay, but extracting a reliable gene-level result requires several specialised tools, consistent coordinates and explicit statements about where the data are not sufficient. Openly available workflows such as BRCA-analyzer [5] were developed for amplicon panels on GRCh37/hg19, depend on licence-restricted annotation and do not report copy-number or structural variants. General-purpose germline pipelines such as Sarek [6] cover WGS but are not tailored to a concise, gene-focused report with exon-level coverage and conservative interpretation hints.

brca12-wgs addresses this gap with three design goals: (i) GRCh38 only, with automatic rejection of other assemblies; (ii) separate, traceable evidence layers for small variants, copy-number candidates, structural variants and coverage; and (iii) reproducibility through pinned software environments, recorded tool versions and a verbatim command log.

2 Implementation

The pipeline is written in Nextflow DSL2 [7] and runs locally with conda environments or with tools on the system path (Figure 1, Table 1).

Table 1. Processes, tools and outputs.

Step	Tool (pinned version)	Purpose
Input check	Python	GRCh38 detection from chr13/chr17 lengths; region and exon BED harmonisation
Alignment	BWA-MEM 0.7.18 [8], samtools 1.21 [9]	Alignment, fixmate, sort, duplicate marking
SNVs/indels	GATK 4.6.1.0 HaplotypeCaller [10] or DeepVariant [11]	Calling in both genes ± 200 bp
Structural variants	Manta 1.6.0 [12]	Breakpoints within ± 1 Mb of both genes
Copy number	CNVkit 0.9.11 [13]	WGS mode, flat reference over chr13 and chr17; per-exon depth ratios
Annotation	Ensembl VEP 113 [14], bcftools	MANE Select [15], HGVS, gnomAD [16], ClinVar [17]
Coverage	mosdepth 0.3.10 [18]	Per exon at MAPQ ≥ 20 and ≥ 0; 10/20/30×
Report	Python (standard library)	HTML, Markdown (English or German), JSON

2.1 Reference and regions

The input check reads the FASTA index and accepts the reference only if chr13 and chr17 have the GRCh38 lengths; GRCh37 is rejected with an explicit message. Gene loci are fixed to chr17:43,044,295–43,170,327 (*BRCA1*) and chr13:32,315,077–32,400,268 (*BRCA2*). Exons are taken from the MANE Select transcripts [15] NM_007294.4 (23 exons) and NM_000059.4 (27 exons). For *BRCA1* the report also shows the legacy exon numbering, which omits exon 4.

2.2 Variant calling

Reads are aligned with BWA-MEM [8]; sorting, mate fixing and duplicate marking use samtools [9]. Small variants are called with GATK HaplotypeCaller [10] restricted to the padded gene regions, or optionally with DeepVariant [11]. Manta [12] is run with call regions extending 1 Mb around both genes, so that rearrangements with one breakpoint inside a gene are retained, including breakend (BND) pairs. Two breakends a few bases apart whose mates lie elsewhere are flagged as a possible insertion of foreign sequence such as a mobile element. In addition, soft-clipped read ends inside the genes are searched for the signature of an insertion: reads from the left clipped at one position, reads from the right clipped at a position up to 30 bp before it (the target-site duplication), at least three clipped reads on each side and a clipped fraction of at least 15 %. Such a pair is listed as an insertion signature when the clipped sequence contains k-mers of the Alu consensus; it is a hint from the reads, not a variant call. For copy number, CNVkit [13] is run in WGS mode with 1 kb bins over chr13 and chr17 and a flat reference; bins with almost no coverage are excluded from segmentation (Section 3.3). Segments overlapping a gene with an estimated copy number different from two are reported as candidates with the affected exons. Independently, the mean depth of each exon is divided by the median depth of 100 kb windows on the same chromosome; ratios below 0.65 or above 1.35 are listed as dosage hints. Because only a single sample is available, both are labelled as candidates that require MLPA [19] or another validated method.

2.3 Annotation and ACMG/AMP hints

VEP [14] annotates consequences, HGVS notation and exon or intron number on the MANE Select transcript. ClinVar [17] significance, review status and condition come from a user-supplied ClinVar VCF, so that the ClinVar version is under the user's control. The pipeline then evaluates single ACMG/AMP criteria [20] following version 1.2 of the ClinGen ENIGMA BRCA1/BRCA2 specification [21]: BA1, BS1 and BS1_Supporting from gnomAD [16] non-cancer filtering allele frequencies in non-founder populations, PM2_Supporting for substitutions absent from both gnomAD data sets, PP3 and BP4 from BayesDel [22] and SpliceAI [23] inside the defined functional domains, BP1_Strong outside them and BP7 for silent and deep intronic variants. The gnomAD, SpliceAI and BayesDel files are supplied by the user; without them the corresponding hints are reported as open or as approximations. Truncating and canonical splice variants are reported as PVS1 candidates, restricted in the last exon [24]; the exon-specific PVS1 decision tree is not yet applied. No combination rule is used and no classification is assigned; the JSON field `clinicalClassification` is always null.

2.4 Coverage and uncertain regions

Depth per exon is computed with mosdepth [18]. An exon is flagged as uncertain if its mean depth is below 20×, if fewer than 95 % of its bases reach 20×, or if more than 10 % of reads have mapping quality below 20, which identifies regions affected by pseudogene homology or repeats. All intervals below 10× are listed. Small variants receive quality flags for non-PASS filters, low depth, low genotype quality, heterozygous allele fractions outside 25–75 %, low mapping quality and location in an uncertain exon. All thresholds are parameters.

2.5 Reporting and reproducibility

The report has separate sections for *BRCA1* and *BRCA2* small variants, copy-number candidates, structural variants, coverage per exon and uncertain intervals, followed by tool versions, the invocation and limitations. It is produced as HTML and Markdown in English or German and as JSON (schema `genome.brca12.v1`). Tool versions are captured at run time, every task copies its command script into `brca12_commands.sh`, and conda environments pin all tool versions. Unit tests and a stub run of the complete workflow run in GitHub Actions on every change.

3 Validation

Four data sets were used (Figures 2 and 3). All scripts, sample lists and result files are in the `validation/` directory of the repository.

3.1 Genome in a Bottle truth sets

Small-variant calls were compared with the GIAB NIST v4.2.1 GRCh38 truth sets for HG001 and HG002 [25, 26]. Illumina HiSeq 2×148 bp read pairs that the GIAB alignments placed within either gene ±10 kb were extracted, downsampled to about 36× and 30×, converted back to FASTQ and processed from FASTQ. Calls were matched with RTG Tools `vcfeval` [27], which compares haplotypes and requires the genotype to agree, inside the GIAB high-confidence regions (91.9 % and 95.1 % of the 211,225 bp of both loci). The pipeline recovered 364 of 364 SNVs and 67 of 69 indels, and all 34 variants in the exons ±20 bp without a false positive. The three discordant calls were intronic; two were genotype errors in homopolymers at 12× and 14× that carried the pipeline's own quality flags.

3.2 Real carriers of pathogenic variants

The 30× joint call set of the 3,202 samples of the 1000 Genomes Project [28] was searched for loss-of-function variants in the MANE coding sequence and its canonical splice sites. Ten individuals carry eight such variants, seven of them annotated as pathogenic in the ClinVar release used by that call set (Table 2). Read pairs over both genes ±10 kb were taken from the public CRAM files and processed from FASTQ. All 10 variants were called heterozygous with genotype quality 99, including a 5 bp deletion that the 1000 Genomes call set itself marks as

low confidence. As a secondary check, calls in the exons ±20 bp agreed with the 1000 Genomes genotypes of the same individuals for 141 of 141 SNVs and 19 of 21 indels, with 2 and 2 additional calls; that call set is a comparison, not a truth set.

Table 2. Carriers of pathogenic *BRCA1*/*BRCA2* variants in the 1000 Genomes Project and the result of the pipeline. AD, reads supporting reference and variant allele.

Sample	Pop.	Gene	Variant (MANE)	Consequence	Detected, genotype	Depth	AD
HG02164	CDX	*BRCA1*	c.34C>T	p.Gln12Ter, nonsense	yes, 0/1	47	23,24
HG03929	BEB	*BRCA1*	c.3403C>T	p.Gln1135Ter, nonsense	yes, 0/1	26	14,12
HG00365	FIN	*BRCA1*	c.4357+1G>A	canonical splice donor	yes, 0/1	33	16,17
NA19397	LWK	*BRCA2*	c.2279_2283del	frameshift	yes, 0/1	25	10,15
HG02274	PEL	*BRCA2*	c.5146_5149del	frameshift	yes, 0/1	25	20,5
NA19130	YRI	*BRCA2*	c.5241_5242insTA	frameshift	yes, 0/1	36	21,15
NA19132	YRI	*BRCA2*	c.5241_5242insTA	frameshift	yes, 0/1	28	9,19
HG01377	CLM	*BRCA2*	c.7063G>T	p.Glu2355Ter, nonsense	yes, 0/1	32	20,12
HG00611	CHS	*BRCA2*	c.7697dup	frameshift	yes, 0/1	46	27,19
HG00612	CHS	*BRCA2*	c.7697dup	frameshift	yes, 0/1	25	20,5

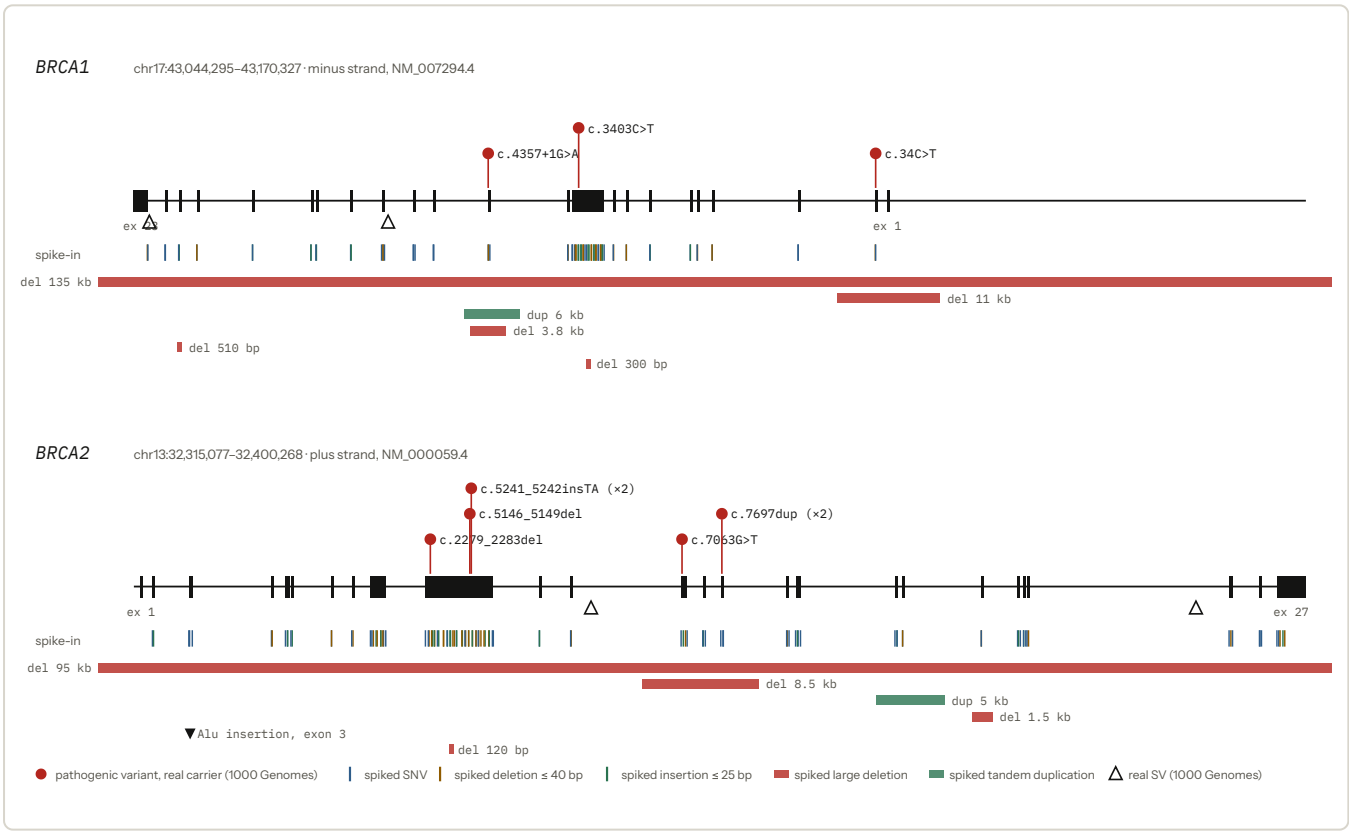


Figure 2. Variants used for validation, drawn on both gene loci. Black boxes are MANE Select exons. Above the gene: pathogenic variants of real carriers. Below: spiked small variants (one tick each), spiked large events and the positions of real structural variants.

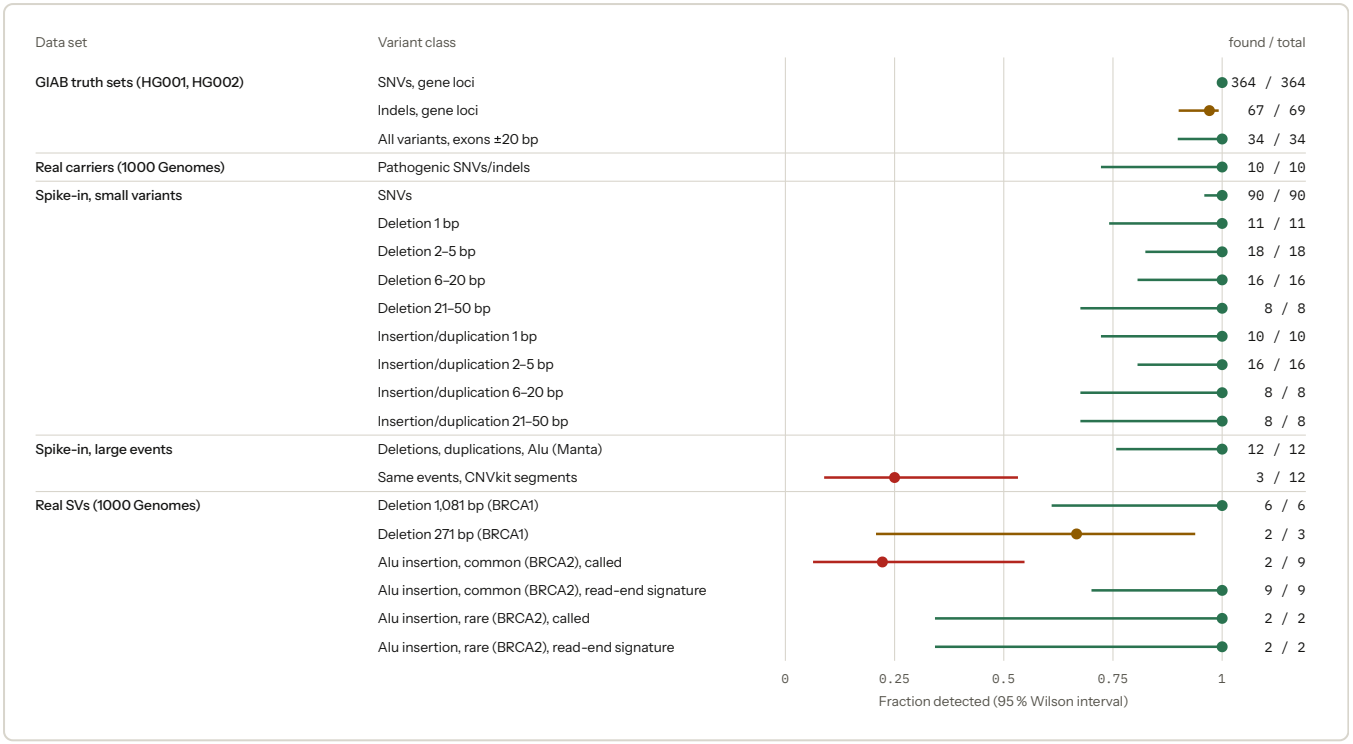


Figure 3. Detection by data set and variant class. Points are the fraction of known variants that the pipeline reported; bars are 95 % Wilson intervals [29]. Green, all found; amber and red, partly found. Small numbers give wide intervals.

3.3 Spike-in experiment

No openly available short-read WGS data set of a carrier of a pathogenic exon deletion, duplication or Alu insertion was found; the GeT-RM reference materials [30] are distributed as DNA. We therefore spiked synthetic heterozygous variants into real reads. The background was GIAB HG002 on chr13 and chr17 at about 30×, which gives CNVkit a real chromosome-wide baseline. Inside both genes ±20 kb every real read pair was replaced by a pair with the same position, fragment length, read lengths and base qualities, read from one of two haplotypes: the reference, or the reference with the spiked events. Sequencing errors were drawn from the real base qualities. Coverage profile and fragment sizes therefore stayed those of the real run. The replaced pairs were aligned with BWA-MEM and merged with the untouched background, and the pipeline was run with real Manta and CNVkit.

Two samples carried 185 small variants: one per coding exon segment with rotating types, SNVs at canonical splice sites, and five variants modelled on well-known pathogenic alleles (*BRCA1* c.68_69del, c.181T>G, c.5266dup; *BRCA2* c.771_775del, c.5946del), whose coordinates were derived from a coding-sequence model that is checked by translation. All 185 were called with the correct genotype, with no false positive in the exons (Figure 3). Outside the exons, every sample including the negative control showed the same two calls at one intronic *BRCA2* site, which come from real HG002 reads with an unmapped mate that the replacement does not cover, and the samples with large deletions showed short artefactual calls directly at some breakpoints.

Six samples carried 12 large events from 120 bp to a whole-gene deletion, two tandem duplications and an Alu insertion in *BRCA2* exon 3 (Figure 4). Manta reported 12 of 12 with exact breakpoints and heterozygous genotype; the Alu insertion appeared as a pair of breakends. The read-end signature (Section 2.2) listed the spiked Alu insertion and nothing in the other eight spike-in samples. With the settings of version 1.0.2, CNVkit reported 0 of 12: bins without coverage dominated the segmentation statistics, and even whole-gene deletions with a clear half-depth signal were merged into their neighbours (Figure 5). Excluding such bins (`--drop-low-coverage`) raised this to 3 of 12; the smallest event reported by CNVkit was 11 kb. In addition, exon dosage hints had been suppressed for exactly the affected exons, because a heterozygous deletion also makes an exon fall below the depth threshold. After the correction, 9 of the 11 deletions and duplications produced a dosage hint for an affected exon. A third defect stopped the pipeline when the input BAM was named like the sample.

Spiked event (heterozygous)		Size	Manta	CNVkit 1.0.2	CNVkit 1.1.0	Exon depth hint	Small-variant caller
BRCA2 Deletion	120 bp deletion inside exon 11	120 bp	✓	○	○	○	✓
BRCA1 Deletion	300 bp deletion inside exon 10	300 bp	✓	○	○	○	○
BRCA2 Alu insertion	exon 3 between c.156 and c.157	327 bp	✓	n/a	n/a	n/a	○
BRCA1 Deletion	exon 21	510 bp	✓	○	○	✓	n/a
BRCA2 Deletion	exon 21	1.5 kb	✓	○	○	✓	n/a
BRCA1 Deletion	exon 12	3.8 kb	✓	○	○	✓	n/a
BRCA2 Tandem duplication	exons 19 and 20	5 kb	✓	○	○	✓	n/a
BRCA1 Tandem duplication	exon 12	6 kb	✓	○	○	✓	n/a
BRCA2 Deletion	exons 14 to 16	8.5 kb	✓	○	○	✓	n/a
BRCA1 Deletion	exons 1 and 2 including the promoter	11 kb	✓	○	✓	✓	n/a
BRCA2 Deletion	the whole gene	95 kb	✓	○	✓	✓	n/a
BRCA1 Deletion	the whole gene	135 kb	✓	○	✓	✓	n/a
Detected			12 / 12	0 / 12	3 / 12	9 / 12	1 / 12

Filled: reported with the correct type at the spiked position. Open: not reported. Size bar on a log scale.

Figure 4. Spiked large events and the evidence layer that reported them. CNVkit 1.0.2 and 1.1.0 differ only in the exclusion of bins without coverage. Single-sample depth methods are not expected to resolve events of one or a few 1 kb bins; n/a marks layers that cannot apply.

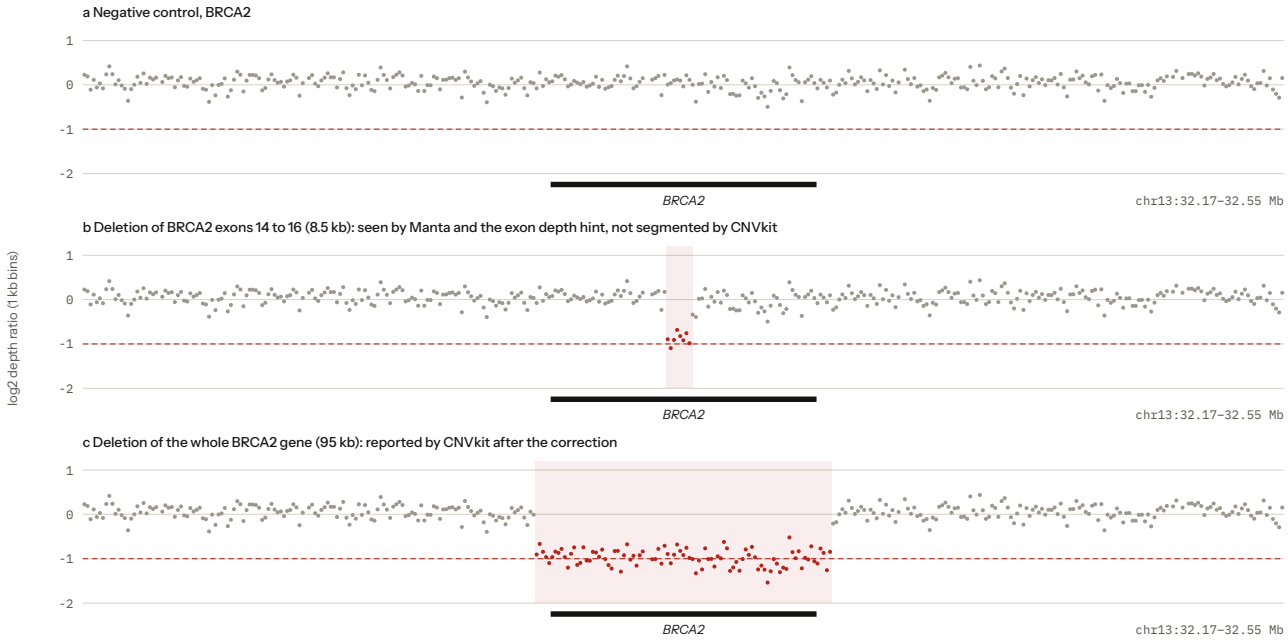


Figure 5. Read-depth signal of spiked heterozygous deletions. Each point is a 1 kb bin (CNVkit log2 ratio). The dashed line marks the expected value for one copy (−1). Shaded: spiked interval. The signal is present in the bins; whether it is reported depends on the segmentation.

3.4 Real structural variants

The phased SNV, indel and SV panel of the 1000 Genomes Project [28] contains two Alu insertions in *BRCA2* and two deletions in *BRCA1*, all intronic or otherwise not disease-causing. They allow the same kind of event to be tested in real reads (Table 3). Carriers and six samples that are homozygous reference at all four sites were processed as in Section 3.2, with one addition: mates that the original alignment had placed elsewhere in the genome were retrieved as well, because read pairs with a mate inside the inserted element are otherwise lost: without them only 2 of 11 Alu insertions were called and 9 of 11 showed the read-end signature. At the two Alu sites the panel genotypes were looked up for all 33 samples of Sections 3.2 and 3.4, which added three heterozygous carriers of the common insertion; the ten samples of Section 3.2 were rerun with retrieved mates for this purpose. The panel calls are statistically phased short-read calls (the Alu insertions come from MELT [31] within the GATK-SV pipeline [32]) and serve as a comparison, not as truth.

Table 3. Real structural variants in 1000 Genomes samples. Called: by Manta or the small-variant caller at the site. Signature: insertion signature from clipped read ends. Controls: samples with panel genotype 0|0 and any call or signature at the site.

Gene	Event	Position	Panel caller	Called	Signature	Controls
BRCA2	Alu insertion, 281 bp	chr13:32,348,302	MELT (GATK-SV)	2 / 9	9 / 9	0 / 22
BRCA2	Alu insertion, 281 bp	chr13:32,392,272	MELT (GATK-SV)	2 / 2	2 / 2	0 / 25
BRCA1	Deletion, 271 bp	chr17:43,045,958	svtools	2 / 3	n/a	0 / 6
BRCA1	Deletion, 1,081 bp	chr17:43,071,626	Manta, Wham (GATK-SV)	6 / 6	n/a	0 / 6

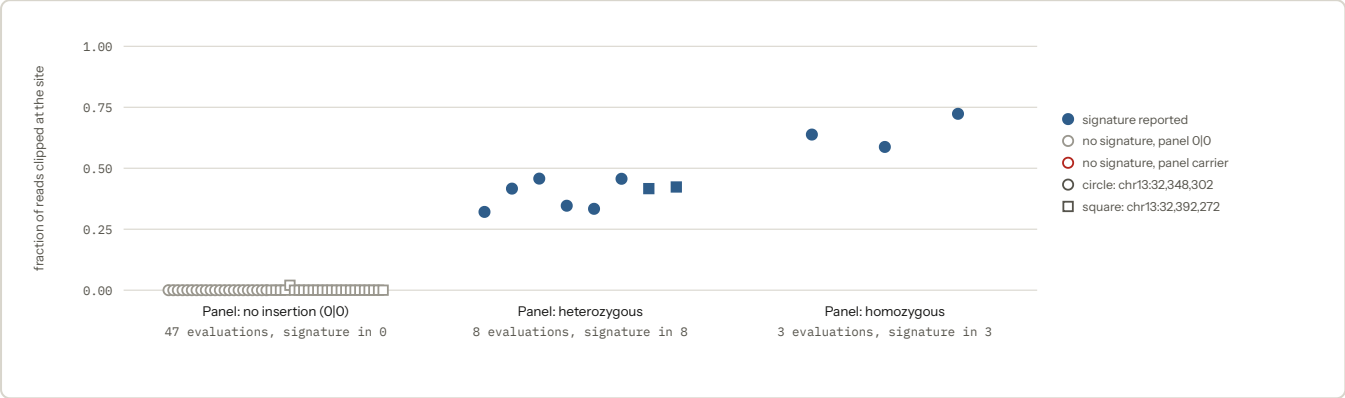


Figure 6. Read ends at the two real Alu insertion sites. Each symbol is one sample at one site: the fraction of reads over the site that are clipped there by at least 10 bases, grouped by the genotype in the 1000 Genomes panel. Filled: the pipeline listed an insertion signature with Alu sequence. The raw fraction counts every read within 15 bp of the site and is therefore lower than the clipped fraction computed by the detector.

The 1,081 bp deletion was reported by Manta in 6 of 6 carriers, with the genotype of the panel including the homozygous carrier; its start lies 101 bp from the panel coordinate, whose breakpoints are not base-exact. The 271 bp deletion was reported in 2 of 3 carriers. In the carrier without a call (HG01796) the depth inside the deleted interval equals the depth next to it (32x and 33x; 13x and 31x in the carrier HG02601), so the reads of this sample do not show the deletion. The Alu insertions were called in 2 of 9 carriers of the common and 2 of 2 carriers of the rare insertion, by Manta as insertions (about 320 bp where the length was resolved) or by the small-variant caller as a partial insertion. The read-end signature was present in 9 of 9 and 2 of 2 carriers, with clipped fractions of 0.43 to 0.64 in heterozygous and 0.92 to 0.94 in homozygous carriers according to the panel. No sample with panel genotype 0|0 had a call or signature at any of the four sites.

3.5 Limits of this validation

- Reads for Sections 3.1, 3.2 and 3.4 were taken from existing alignments near the genes and realigned against chr13 and chr17 only. This removes competition from paralogous sequence elsewhere, which may overstate precision. A whole-genome run against the complete reference is the next step.
- Spiked variants are synthetic. They test whether a variant that is present in the reads is called and reported, not how real pathogenic rearrangements look. The spiked Alu element was copied from the reduced reference, which makes its detection easier than a real mobile element insertion.
- No Ensembl VEP cache was available on the validation machine, so annotation and the ACMG/AMP hints were not part of these runs. In Sections 3.1, 3.2 and 3.4 CNVkit was replaced by a neutral stand-in.
- Numbers are small. Ten carriers and twelve large events give wide confidence intervals (Figure 3).
- The read-end signature was written after the missed Alu insertions had been inspected, so its result on these carriers is not an independent test.
- No real sample with a pathogenic exon deletion, duplication or Alu insertion was tested.

4 Comparison with existing workflows

Table 4. Feature comparison with BRCA-analyzer [5].

	brca12-wgs	BRCA-analyzer
Data type	WGS (FASTQ or BAM/CRAM)	Amplicon panels
Assembly	GRCh38, enforced	hg19
Small variants	GATK 4 HaplotypeCaller or DeepVariant	GATK 3.6/3.7
Copy number / SV	CNVkit + exon dosage / Manta	Not included
Annotation	VEP, MANE Select, gnomAD, user ClinVar	ANNOVAR, snpEff, ClinVar 2018
Interpretation support	ACMG/AMP criterion hints (ENIGMA v1.2), no classification	Not included
Published validation	GIAB, real carriers, spike-ins	Not stated
Licence	MIT	GPL-3.0

5 Limitations

- Short-read WGS has limited sensitivity for mosaic variants below about 15 % allele fraction, for complex rearrangements and in regions with pseudogene homology such as *BRCA1* exons 1–2.
- Copy-number results come from a single sample with a flat reference and are candidates only. In the spike-in experiment CNVkit alone missed every event below 11 kb; detection of exon-level events currently rests on Manta, which needs breakpoints inside mappable sequence.
- Mobile element insertions have no dedicated caller. Manta missed real Alu insertions in several carriers; the read-end signature recognises Alu sequence only, was developed on the same two insertion sites it is reported for here, and has not been tested on an independent set or on L1 and SVA elements (Section 3.4).
- The ACMG/AMP hints cover a subset of criteria. The PVS1 decision tree, functional (PS3/BS3), segregation (PP1), case-control (PS4) and same-codon (PS1/PM5) evidence are not evaluated.
- The pipeline has not been validated for clinical use.

6 Availability and requirements

Repository: https://github.com/pjlabsdev/brca12-wgs · **Archived version:** DOI 10.5281/zenodo.24173638 (version 1.0.1) · **Operating systems:** Linux, macOS · **Requirements:** Nextflow ≥ 23.10, Java 17, conda/mamba or the listed tools, GRCh38 FASTA with index and dictionary, Ensembl VEP GRCh38 cache, ClinVar VCF · **Licence:** MIT.

```
nextflow run main.nf -profile conda \
  --sample S --reads_r1 S_R1.fastq.gz --reads_r2
S_R2.fastq.gz \
  --reference GRCh38.fa --vep_cache ~/.vep --
clinvar_vcf clinvar.vcf.gz
```


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Conflict of interest

None declared.

References

1. Kuchenbaecker KB, Hopper JL, Barnes DR, et al. Risks of breast, ovarian, and contralateral breast cancer for BRCA1 and BRCA2 mutation carriers. JAMA. 2017;317(23):2402–2416.
2. Sluiter MD, van Rensburg EJ. Large genomic rearrangements of the BRCA1 and BRCA2 genes: review of the literature and report of a novel BRCA1 mutation. Breast Cancer Res Treat. 2011;125(2):325–349.
3. Teugels E, De Brakeleer S, Goelen G, Lissens W, Sermijn E, De Grève J. De novo Alu element insertions targeted to a sequence common to the BRCA1 and BRCA2 genes. Hum Mutat. 2005;26(3):284.
4. Machado PM, Brandão RD, Cavaco BM, et al. Screening for a BRCA2 rearrangement in high-risk breast/ovarian cancer families: evidence for a founder effect and analysis of the associated phenotypes. J Clin Oncol. 2007;25(15):2027–2034.
5. Kechin A. BRCA-analyzer: automatic workflow for analysis of BRCA1/2 NGS data. https://github.com/aakechin/BRCA-analyzer (accessed 12 November 2026).
6. Garcia M, Juhos S, Larsson M, et al. Sarek: a portable workflow for whole-genome sequencing analysis of germline and somatic variants. F1000Research. 2020;9:63.
7. Di Tommaso P, Chatzou M, Floden EW, et al. Nextflow enables reproducible computational workflows. Nat Biotechnol. 2017;35:316–319.
8. Li H. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. arXiv:1303.3997; 2013.
9. Danecek P, Bonfield JK, Liddle J, et al. Twelve years of SAMtools and BCFtools. GigaScience. 2021;10(2):giab008.
10. Poplin R, Ruano-Rubio V, DePristo MA, et al. Scaling accurate genetic variant discovery to tens of thousands of samples. bioRxiv 201178; 2018.
11. Poplin R, Chang PC, Alexander D, et al. A universal SNP and small-indel variant caller using deep neural networks. Nat Biotechnol. 2018;36:983–987.
12. Chen X, Schulz-Trieglaff O, Shaw R, et al. Manta: rapid detection of structural variants and indels for germline and cancer sequencing applications. Bioinformatics. 2016;32(8):1220–1222.
13. Talevich E, Shain AH, Botton T, Bastian BC. CNVkit: genome-wide copy number detection and visualization from targeted DNA sequencing. PLoS Comput Biol. 2016;12(4):e1004873.
14. McLaren W, Gil L, Hunt SE, et al. The Ensembl Variant Effect Predictor. Genome Biol. 2016;17:122.
15. Morales J, Pujar S, Loveland JE, et al. A joint NCBI and EMBL-EBI transcript set for clinical genomics and research. Nature. 2022;604:310–315.

16. Karczewski KJ, Francioli LC, Tiao G, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. Nature. 2020;581:434–443.
17. Landrum MJ, Lee JM, Benson M, et al. ClinVar: improving access to variant interpretations and supporting evidence. Nucleic Acids Res. 2018;46(D1):D1062–D1067.
18. Pedersen BS, Quinlan AR. Mosdepth: quick coverage calculation for genomes and exomes. Bioinformatics. 2018;34(5):867–868.
19. Schouten JP, McElgunn CJ, Waaijer R, Zwijnenburg D, Diepvens F, Pals G. Relative quantification of 40 nucleic acid sequences by multiplex ligation-dependent probe amplification. Nucleic Acids Res. 2002;30(12):e57.
20. Richards S, Aziz N, Bale S, et al. Standards and guidelines for the interpretation of sequence variants. Genet Med. 2015;17(5):405–424.
21. Parsons MT, de la Hoya M, Richardson ME, et al. Evidence-based recommendations for gene-specific ACMG/AMP variant classification from the ClinGen ENIGMA BRCA1 and BRCA2 Variant Curation Expert Panel. Am J Hum Genet. 2024;111(9):2044–2058.
22. Feng BJ. PERCH: a unified framework for disease gene prioritization. Hum Mutat. 2017;38(3):243–251.
23. Jaganathan K, Kyriazopoulou Panagiotopoulou S, McRae JF, et al. Predicting splicing from primary sequence with deep learning. Cell. 2019;176(3):535–548.
24. Abou Tayoun AN, Pesaran T, DiStefano MT, et al. Recommendations for interpreting the loss of function PVS1 ACMG/AMP variant criterion. Hum Mutat. 2018;39(11):1517–1524.
25. Zook JM, Chapman B, Wang J, et al. Integrating human sequence data sets provides a resource of benchmark SNP and indel genotype calls. Nat Biotechnol. 2014;32:246–251.
26. Wagner J, Olson ND, Harris L, et al. Benchmarking challenging small variants with linked and long reads. Cell Genomics. 2022;2(5):100128.
27. Cleary JG, Braithwaite R, Gaastra K, et al. Comparing variant call files for performance benchmarking of next-generation sequencing variant calling pipelines. bioRxiv 023754; 2015.
28. Byrska-Bishop M, Evani US, Zhao X, et al. High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios. Cell. 2022;185(18):3426–3440.
29. Wilson EB. Probable inference, the law of succession, and statistical inference. J Am Stat Assoc. 1927;22(158):209–212.
30. Barker SD, Bale S, Booker J, et al. Development and characterization of reference materials for MTHFR, SERPINA1, RET, BRCA1, and BRCA2 genetic testing. J Mol Diagn. 2009;11(6):553–561.
31. Gardner EJ, Lam VK, Harris DN, et al. The Mobile Element Locator Tool (MELT): population-scale mobile element discovery and biology. Genome Res. 2017;27(11):1916–1929.
32. Collins RL, Brand H, Karczewski KJ, et al. A structural variation reference for medical and population genetics. Nature. 2020;581:444–451.